

Supplementary material

How reliable is image-based diagnosis of durian (*Durio zibethinus*) diseases from field photographs? Grouped cross-validation, run-to-run variation and transfer to another country

Lin Ding Shan (Independent researcher, Malaysia)

All tables and figures are generated by `paper/make_supplement.py` from the released result files (`results/preds.csv`, `results/results.csv`, `results/v2/results.csv`, `results/numbers.json` and `paper/numbers.json`). The single-channel spatial gate (SG) is labelled `lfa` / LFA in the code, the result files and Figs. S1–S2, and the leaf blight complex is labelled `Leaf_rot`.

Table S1. Per-class precision, recall and F1 (% , mean $\pm$ SD over runs in which the class was present), all test images

Class	Configuration	Precision	Recall	F1
Algal leaf spot	None	72.9 $\pm$ 25.5	63.7 $\pm$ 18.4	65.3 $\pm$ 19.3
	SG	73.8 $\pm$ 25.9	61.1 $\pm$ 21.2	63.6 $\pm$ 21.0
	SE	72.8 $\pm$ 25.2	59.9 $\pm$ 20.6	62.5 $\pm$ 19.9
	CBAM	74.1 $\pm$ 24.1	60.7 $\pm$ 21.8	63.6 $\pm$ 21.1
Leaf blight complex	None	66.6 $\pm$ 15.4	78.9 $\pm$ 16.0	70.7 $\pm$ 12.3
	SG	66.9 $\pm$ 16.2	82.9 $\pm$ 15.1	71.8 $\pm$ 11.1
	SE	64.8 $\pm$ 17.9	79.3 $\pm$ 20.1	69.4 $\pm$ 15.5
	CBAM	66.0 $\pm$ 14.5	82.2 $\pm$ 12.3	71.8 $\pm$ 9.9
<i>Phomopsis</i> leaf spot	None	77.2 $\pm$ 20.5	78.6 $\pm$ 19.0	74.2 $\pm$ 13.4
	SG	77.7 $\pm$ 20.2	76.7 $\pm$ 19.4	73.5 $\pm$ 13.9
	SE	78.3 $\pm$ 20.9	78.9 $\pm$ 18.8	74.9 $\pm$ 14.7
	CBAM	77.0 $\pm$ 21.0	78.1 $\pm$ 20.4	73.4 $\pm$ 14.9
Root and collar rot	None	91.0 $\pm$ 18.3	99.3 $\pm$ 2.1	93.6 $\pm$ 14.2
	SG	92.3 $\pm$ 16.2	99.0 $\pm$ 2.2	94.6 $\pm$ 11.6
	SE	91.2 $\pm$ 18.3	99.3 $\pm$ 1.5	93.8 $\pm$ 14.2
	CBAM	91.5 $\pm$ 17.3	99.2 $\pm$ 2.0	94.0 $\pm$ 13.1

Table S2. Accuracy (% , mean $\pm$ SD over 40 runs) and paired difference from the plain network over 20 folds (corrected resampled *t*-test)

Configuration	Accuracy	Difference (pp) [95% CI]	<i>p</i> , corrected	Folds higher
None	75.95 $\pm$ 9.33	–	–	–
SG	75.47 $\pm$ 10.09	–0.49 [–4.67, 3.70]	0.811	9/20
SE	74.75 $\pm$ 10.93	–1.20 [–8.79, 6.39]	0.744	10/20
CBAM	75.78 $\pm$ 9.66	–0.17 [–4.82, 4.47]	0.939	10/20

Table S3. Preliminary experiment: eight runs per configuration on one fixed five-class split (earlier code; training randomness not controlled by the seed, so replicates are unpaired)

Configuration	Macro F1, mean $\pm$ SD	Difference from none (pp)	p , Welch t	p , Mann–Whitney
None	77.55 $\pm$ 1.91	–	–	–
SG	78.05 $\pm$ 2.61	+0.50	0.671	0.958
SE	76.63 $\pm$ 3.75	–0.92	0.551	0.752
CBAM	77.10 $\pm$ 2.86	–0.45	0.718	0.833

Table S4. Macro F1 averaged over all four classes (an absent class scored as zero), as stored by the training notebooks and used in the pre-specified decision rule

Configuration	Macro F1 (%), mean $\pm$ SD	Paired difference (pp) [95% CI]	p , corrected	Meets rule (2 pp and CI > 0)
None	72.44 $\pm$ 13.05	–	–	–
SG	72.32 $\pm$ 13.08	–0.12 [–4.54, 4.30]	0.954	no
SE	71.63 $\pm$ 13.45	–0.82 [–7.63, 6.00]	0.805	no
CBAM	72.18 $\pm$ 13.43	–0.27 [–5.11, 4.57]	0.909	no

Table S5. Training settings (identical for all configurations)

Item	Setting
Backbone	EfficientNet-B0, torchvision ImageNet weights (IMAGENET1K_V1)
Input	Train: RandomResizedCrop(224, scale 0.7–1.0). Evaluation: Resize(256) on the short side, CenterCrop(224)
Augmentation	Horizontal flip $p = 0.5$; vertical flip $p = 0.3$; RandomAffine (rotation $\pm 20^\circ$, translation 5 %, scale 0.95–1.05); ColorJitter (brightness 0.3, contrast 0.3, saturation 0.2, hue 0.05); GaussianBlur (kernel 3–7, 0.1–1.5); ImageNet normalisation; RandomErasing $p = 0.3$, 2–10 % of the image
Classifier head	Dropout $p = 0.3$, Linear(1280, 4)
Stage 1	15 epochs, backbone frozen (BatchNorm statistics still updating), Adam, lr 1e-3, attention module + head trained
Stage 2	15 epochs, all parameters trainable, Adam, lr 1e-5, starting from the best stage-1 checkpoint
Checkpoint selection	Highest inner-validation macro F1 over both stages; inner validation 10 % of the training part, by whole groups
Loss and sampling	Cross-entropy with inverse-frequency class weights; WeightedRandomSampler with class weights
Batch size	16
Precision	Mixed precision (torch.autocast with a gradient scaler)
Seeds	0 and 1 per fold; Python, NumPy, PyTorch, sampler and data-loader workers seeded; cuDNN deterministic
Hardware	Google Colab GPU runtime; PyTorch and torchvision

Table S6. Paired comparisons of macro F1 with the plain network under three tests over the 20 folds

Configuration	Mean difference (pp)	p , corrected resampled t	p , uncorrected paired t	p , Wilcoxon signed-rank
SG	−0.13	0.953	0.870	0.898
SE	−0.77	0.816	0.521	0.956
CBAM	−0.38	0.879	0.675	0.277

Table S7. Sensitivity to the grouping of video frames (Section 2.3 of the manuscript)

Test images scored	Mean test images per fold	Configuration	Macro F1 (%), mean $\pm$ SD	Paired difference (pp) [95% CI]	p , corrected
All test images	137.5	None	75.06 $\pm$ 10.86	–	–
		SG	74.93 $\pm$ 10.95	−0.13 [−4.56, 4.31]	0.953
		SE	74.29 $\pm$ 11.80	−0.77 [−7.62, 6.08]	0.816
		CBAM	74.68 $\pm$ 10.78	−0.38 [−5.48, 4.73]	0.879
Video frames excluded	121.8	None	66.60 $\pm$ 12.04	–	–
		SG	66.46 $\pm$ 12.15	−0.13 [−4.55, 4.28]	0.950
		SE	65.34 $\pm$ 12.74	−1.26 [−10.28, 7.76]	0.773
		CBAM	66.37 $\pm$ 13.06	−0.23 [−5.83, 5.37]	0.933
Only images whose video-linked group was not split	87.0	None	67.36 $\pm$ 12.17	–	–
		SG	66.54 $\pm$ 12.34	−0.83 [−6.09, 4.44]	0.746
		SE	64.54 $\pm$ 14.16	−2.83 [−11.70, 6.04]	0.512
		CBAM	66.19 $\pm$ 14.53	−1.17 [−7.96, 5.62]	0.722

Without the video frames the mean seed difference was 5.67 points, and the SG was at least 3 points ahead of the plain network in 20.0 % of seed-matched pairs and at least 3 points behind in 22.5 %.

Table S8. Model size, computation and CPU latency for one 224×224 image (median and 90th percentile of 200 passes, Colab CPU)

Configuration	Parameters (M)	Added parameters	GMACs	1 thread, median (ms)	1 thread, p90 (ms)	12 threads, median (ms)	12 threads, p90 (ms)
None	4.0127	0	0.3845	34.29	35.38	22.46	46.40
SG	4.0140	1,281	0.3846	34.18	35.39	22.65	23.92
SE	4.2175	204,800	0.3847	34.21	35.17	22.54	24.78
CBAM	4.2176	204,898	0.3850	35.20	35.86	23.29	27.03

Table S9. Fold sizes and macro F1 per fold (% , mean of two seeds; † no root and collar rot image in the test fold)

Repeat	Fold	Train	Validation	Test	None	SG	SE	CBAM
0	0	440	58	52	94.40	93.28	88.25	95.07
0	1	386	41	123	75.76	77.22	78.18	75.87
0	2	275	28	247	63.87	58.34	62.51	61.07
0	3	388	34	128	82.14	81.89	77.97	81.90
1	0	453	42	55	79.10	71.19	75.34	71.92
1	1	395	46	109	76.57	81.34	80.37	75.90
1	2†	296	30	224	49.82	50.38	50.71	49.08
1	3	348	40	162	74.04	74.75	73.69	73.77
2	0†	419	51	80	72.23	70.65	72.16	72.86
2	1	256	35	259	66.89	68.65	68.43	66.97
2	2	443	46	61	82.37	81.21	84.31	80.29
2	3	358	42	150	81.42	75.09	80.30	80.01
3	0	393	61	96	69.43	74.90	50.39	80.74
3	1	424	46	80	70.07	70.00	70.57	69.69
3	2	247	23	280	69.88	71.23	70.92	67.90
3	3	417	39	94	87.54	87.58	88.28	88.29
4	0†	459	50	41	87.21	88.04	89.95	78.82
4	1	381	34	135	82.71	80.84	82.08	81.72
4	2	309	31	210	56.47	59.59	64.50	61.42
4	3	342	44	164	79.27	82.45	76.84	80.39

Figures S1 and S2

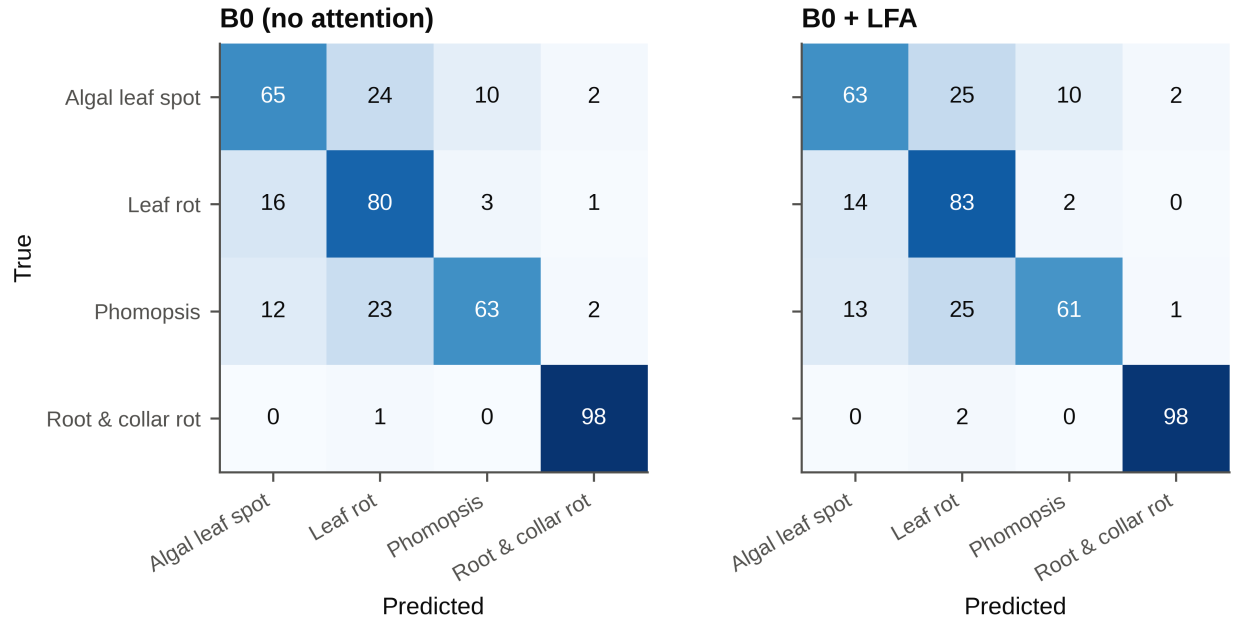


Figure S1: Confusion matrices of the plain network and the SG (labelled LFA), pooled over the 40 runs of each configuration and normalised by row (%), all test images.

Table S10. Per-class precision, recall and F1 (% , mean $\pm$ SD over runs in which the class was present), video frames excluded (primary scoring)

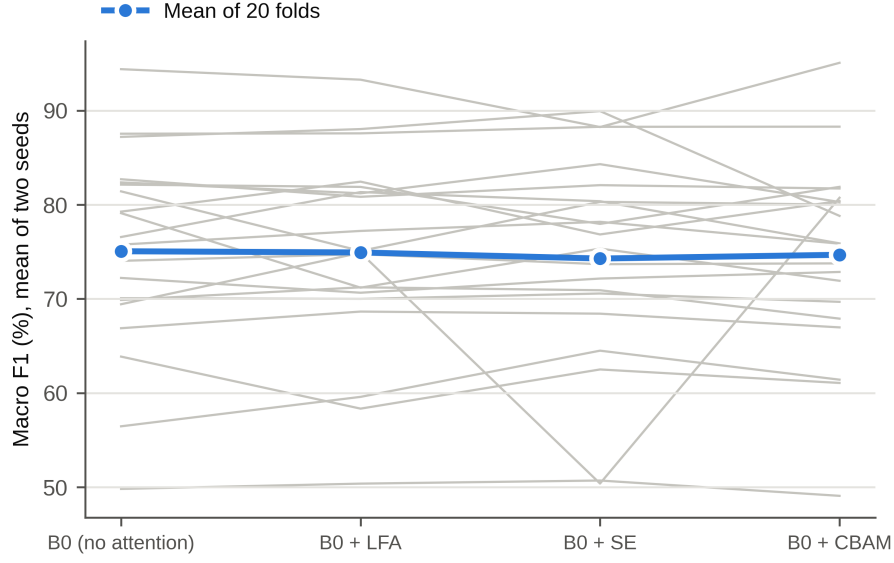


Figure S2: Macro F1 of each configuration on each of the 20 folds (grey lines, mean of two seeds) and the mean over folds (blue), all test images.

Class	Configuration	Precision	Recall	F1	Runs
Algal leaf spot	None	73.1 ± 25.5	63.7 ± 18.4	65.4 ± 19.3	40
	SG	74.1 ± 26.0	61.1 ± 21.2	63.7 ± 21.1	40
	SE	73.1 ± 25.3	59.9 ± 20.6	62.5 ± 19.9	40
	CBAM	74.6 ± 24.4	60.7 ± 21.8	63.7 ± 21.1	40
Leaf blight complex	None	67.5 ± 15.8	78.9 ± 16.0	71.1 ± 12.3	40
	SG	68.0 ± 16.4	82.9 ± 15.1	72.5 ± 11.0	40
	SE	65.9 ± 18.5	79.3 ± 20.1	69.9 ± 15.5	40
	CBAM	66.7 ± 14.7	82.2 ± 12.3	72.2 ± 9.9	40
<i>Phomopsis</i> leaf spot	None	44.3 ± 44.2	29.4 ± 27.8	32.0 ± 31.6	32
	SG	44.4 ± 45.0	27.0 ± 27.3	30.9 ± 31.7	32
	SE	44.1 ± 44.0	31.0 ± 28.2	32.8 ± 32.3	32
	CBAM	44.2 ± 44.4	28.6 ± 28.2	31.4 ± 31.7	32
Root and collar rot	None	91.0 ± 18.3	99.3 ± 2.1	93.6 ± 14.2	34
	SG	92.3 ± 16.2	99.0 ± 2.2	94.6 ± 11.6	34
	SE	91.2 ± 18.3	99.3 ± 1.5	93.8 ± 14.2	34
	CBAM	91.5 ± 17.3	99.2 ± 2.0	94.0 ± 13.1	34